

Envy-Free Chore Allocation with Unit Subsidies under Negative Dichotomous Valuations

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Abstract

We study the allocation of indivisible chores with subsidies under negative dichotomous valuations, where the marginal disutility of every chore is either zero or one. This is the chore analogue of the dichotomous goods model of Barman et al., for which a subsidy of at most one unit per agent is known to suffice for envy-freeness.

We show that the same optimal guarantee holds for chores under no structural assumption beyond binary marginals. For every instance with negative dichotomous valuations, there exists a complete allocation $\mathcal{A}$ and a subsidy vector $p \in \{0, 1\}^n$ with $\sum_{i \in N} p_i \leq n - 1$ such that $(\mathcal{A}, p)$ is envy-free. Moreover, the allocation $\mathcal{A}$ is EF1 even before subsidies are paid. The subsidy bound is tight, and both the allocation and the subsidies can be computed in polynomial time in the value-oracle model. We further show that the unit-subsidy guarantee cannot, in general, be maintained if Pareto optimality is also required.

Unlike in the corresponding goods setting, a natural incremental extension approach does not work for chores. Our proof instead exploits key properties of an envy-free partial allocation and its associated equality graph to obtain the complete allocation and determine which agents require subsidies.

Keywords

fair division, indivisible chores, envy-freeness, subsidies, negative dichotomous valuations, EF1

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1 Introduction

Discrete fair division studies the allocation of indivisible resources among agents with heterogeneous preferences, and lies at the interface of mathematical economics and computer science [4, 8]. A central fairness criterion is *envy-freeness* [9, 19]: no agent prefers another agent’s bundle to her own. For indivisible items, however, an envy-free allocation need not exist, as the standard one-item two-agent instance illustrates. This has motivated a large literature on relaxations of envy-freeness, including envy-freeness up to one good (EF1) [6, 15] and the stronger notion of envy-freeness up to any good (EFX) [7].

A second and complementary line of work retains exact envy-freeness by allowing monetary subsidies. In addition to her allocated bundle A_i , each agent i receives a subsidy $p_i \geq 0$ from a third party, and the outcome is envy-free if

$$v_i(A_i) + p_i \geq v_i(A_j) + p_j \quad \text{for all } i, j.$$

Halpern and Shah [12] show that every instance admits an envy-freeable allocation and study the minimum subsidy required to achieve envy-freeness. Under the standard normalization that each item value is at most one,¹ they show that, for a given envy-freeable allocation, a total subsidy of at most $(n - 1)m$ may be required in the worst case, and conjecture that a total subsidy of $n - 1$ always suffices when the allocation can also be chosen. Brustle et al. [5] prove this conjecture for additive valuations, obtaining an allocation that is also EF1 and balanced, and show that for general monotone valuations a subsidy of at most $2(n - 1)$ per agent suffices.

Barman et al. [2] obtain a substantially stronger guarantee for dichotomous valuations. When every marginal $v_i(S \cup \{g\}) - v_i(S)$ belongs to $\{0, 1\}$, they show that there exists an allocation admitting subsidies in $\{0, 1\}^n$, with total subsidy at most $n - 1$, and that such an outcome can be computed in polynomial time in the value-oracle model. This bound is tight: for n agents and a single good valued at one by every agent, the $n - 1$ agents who do not receive the good each require one unit of subsidy.

In this paper, we consider the corresponding problem for indivisible chores, where receiving an additional item weakly decreases an agent’s utility. Chore-allocation problems arise naturally in applications such as shift assignment, on-call duties, teaching and refereeing loads, committee service, and maintenance tasks. Monetary compensation is also natural in such settings, where it may be interpreted as payment for undertaking an undesirable task or additional workload.

We study *negative dichotomous* valuations: for every agent i , every $S \subseteq M$, and every $g \in M \setminus S$,

$$v_i(S \cup \{g\}) - v_i(S) \in \{0, -1\}.$$

Thus, every item is a chore for every agent, and adding a chore changes an agent’s utility by either zero or minus one. This domain is the sign analogue of the dichotomous goods model of [2].

Binary marginals capture settings in which an additional task either creates a new unit of burden or can be absorbed by an existing commitment. For example, an agent already covering an on-call window may incur no additional cost from another incident during that window, or a driver already routed through a district may absorb an additional stop at no extra cost. Since whether a chore is costly may depend on the rest of the assigned bundle, the model naturally allows non-additive valuations, including valuations that are not subadditive.

¹Such a normalization is necessary when stating absolute bounds on monetary subsidies.



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Although chore allocation is formally related to goods allocation by reversing the direction of preferences, this correspondence does not generally extend to algorithmic arguments. Several works have observed that algorithms and structural results for goods need not carry over directly to chores [1, 3, 17]. The same difficulty arises in our setting. The algorithm of Barman et al. [2] for dichotomous goods is incremental: it allocates the goods one at a time while maintaining an envy-freeable partial allocation whose required subsidies lie in $\{0, 1\}^n$. For chores, one might naturally try to preserve the same invariant by assigning each new chore to an agent who currently receives no subsidy. This approach, however, may not succeed. There are negative dichotomous instances in which a partial allocation admits an envy-free solution with subsidies in $\{0, 1\}^n$, but after introducing one additional chore, no assignment of that chore to an agent preserves the unit-subsidy guarantee; this remains true even if the previously allocated bundles are allowed to be permuted before assigning the new chore. We give such an example in Appendix A. Thus, the one-item extension principle underlying the goods-side construction fails for chores, and a different argument is required.

A general subsidy bound for our setting follows from the work of Kawase et al. [13]. They show that, for doubly monotone valuations with marginal values bounded in absolute value by one, an EF1 allocation can be converted into an envy-free solution with a subsidy of at most $n - 1$ per agent and at most $n(n - 1)/2$ in total. Negative dichotomous valuations form a subclass of this domain. Thus, Kawase et al. [13] already imply, for our setting, a subsidy bound that is independent of the number of chores: at most $n - 1$ per agent and $n(n - 1)/2$ in total. The question is whether this can be strengthened to the unit-subsidy guarantee known for dichotomous goods. Our main result answers this question affirmatively.

THEOREM. *For every instance with negative dichotomous valuations, there exists a complete allocation $\mathcal{A}$ and a subsidy vector $p \in \{0, 1\}^n$ such that $(\mathcal{A}, p)$ is envy-free and*

$$\sum_{i \in N} p_i \leq n - 1.$$

Moreover, $\mathcal{A}$ is EF1 even before subsidies are paid, and such an allocation and subsidy vector can be computed in polynomial time in the value-oracle model.

An additional feature of our result is that the underlying allocation is EF1 even before subsidies are paid. This is not guaranteed by the construction of Barman et al. [2] for dichotomous goods. Thus, beyond extending the unit-subsidy guarantee to negative dichotomous chores, our construction provides an additional fairness guarantee for the allocation itself.

The total subsidy bound is tight, even for identical binary additive valuations. Consider n agents and $n - 1$ chores with

$$v_i(S) = -|S| \quad \text{for every agent } i.$$

In every complete allocation, some agent, say h , receives the empty bundle. Comparing any agent i with agent h in the envy-freeness inequalities gives

$$-|A_i| + p_i \geq v_i(A_h) + p_h = p_h,$$

and hence

$$p_i \geq |A_i| + p_h \geq |A_i|.$$

Therefore,

$$\sum_{i \in N} p_i \geq \sum_{i \in N} |A_i| = n - 1.$$

Thus, the total subsidy bound of $n - 1$ is optimal.

Our proof follows a different route from the incremental construction of Barman et al. [2]. We start from the polynomial-time algorithm of Tao et al. [18], which returns an envy-free partial allocation while leaving at most $n - 1$ chores unassigned. At termination, we assign the remaining chores to distinct agents in a suitable tail strongly connected component of the associated equality graph. We then identify the subsidised agents by tracing equality edges backwards from the agents who receive the remaining chores. The resulting complete allocation is EF1, and assigning one unit of subsidy to the identified agents makes it exactly envy-free.

Finally, the unit-subsidy guarantee cannot, in general, be strengthened to require Pareto optimality. There are instances with two agents and identical binary submodular costs in which every Pareto-optimal allocation requires a subsidy of at least 2 for some agent; see Proposition 3.11.

Section 2 introduces the model and the necessary preliminaries. Section 3 presents the algorithm and proves the main result.

1.1 Related work

The literature on envy-freeness with subsidies has been developed primarily for goods. Halpern and Shah [12] give a characterisation of envy-freeable allocations and obtain the optimal total subsidy bound of $n - 1$ for binary additive valuations. Goko et al. [11] study matroidal valuations and obtain a truthful mechanism achieving envy-freeness and utilitarian optimality with at most one unit of subsidy per agent, under free disposal. If all goods are required to be allocated, they retain the unit-subsidy and efficiency guarantees, but without truthfulness. Barman et al. [2] show that, if truthfulness is not required, no structure beyond binary marginals is needed for the unit-subsidy guarantee. Kawase et al. [13] consider the substantially broader class of doubly monotone valuations and show that any EF1 allocation can be converted into an envy-free solution using at most $n - 1$ units of subsidy per agent and $n(n - 1)/2$ in total. Since negative dichotomous valuations form a subclass of doubly monotone valuations, their result gives the previously known m -independent subsidy benchmark for our domain, but does not yield the unit-subsidy guarantee established here.

Fair allocation of indivisible chores without subsidies has also received considerable attention. Bhaskar et al. [3] give polynomial-time algorithms for EF1 allocations of chores and doubly monotone instances, and show that deciding the existence of an exactly envy-free chore allocation is NP-complete already for binary additive costs. Tao et al. [18] study binary-marginal chores and prove the existence of an EFX and Pareto-optimal allocation for binary additive costs. The additive assumption cannot be dropped in general: Lin et al. [14] construct binary XOS and binary supermodular instances for which no complete EFX allocation exists. In contrast, our result applies throughout the full binary-marginal chore domain: exact envy-freeness is obtained after subsidies, while the underlying allocation is always EF1.

Subsidies have also been studied directly for chore allocation. For proportionality under additive costs, tight bounds are known for

equal entitlements [20], and weighted variants have been studied in [10, 21]. More recently, Lu et al. [16] show that for additive instances containing both goods and chores, one unit of subsidy per agent suffices and the total bound of $n - 1$ is tight. Their result and ours apply to incomparable domains: negative dichotomous valuations need not be additive, while additive valuations need not be dichotomous. Thus, prior to the present work, the unit-subsidy guarantee was known for additive chores [16] and for dichotomous goods [2], but not for non-additive dichotomous chores.

To the best of our knowledge, this is the first unit-subsidy guarantee for envy-freeness under negative dichotomous valuations.

2 Model and Preliminaries

We consider the allocation of a set M of m indivisible chores among a set $N = [n]$ of agents. Each agent $i \in N$ has a valuation $v_i : 2^M \rightarrow \mathbb{R}$ with $v_i(\emptyset) = 0$. Algorithms operate in the standard *value-oracle* model: an oracle returns $v_i(S)$ for any queried set $S \subseteq M$.

An *allocation* $\mathcal{A} = (A_1, \dots, A_n)$ is a tuple of pairwise disjoint bundles, where A_i is assigned to agent i . It is *complete* if $\bigcup_{i \in N} A_i = M$, and *partial* otherwise.

2.1 Negative dichotomous valuations

Definition 2.1 (Negative dichotomous valuation). A valuation v_i is *negative dichotomous* if, for every $S \subseteq M$ and every $e \in M \setminus S$,

$$v_i(S \cup \{e\}) - v_i(S) \in \{-1, 0\}.$$

Thus, adding a chore weakly decreases an agent's value. We impose no further structure; in particular, valuations need not be additive, submodular, or subadditive.

It is convenient to work with the corresponding nonnegative cost function

$$c_i(S) := -v_i(S).$$

Definition 2.2 (Dichotomous cost function). A cost function $c : 2^M \rightarrow \mathbb{R}$ is *dichotomous* if $c(\emptyset) = 0$ and

$$c(S \cup \{e\}) - c(S) \in \{0, 1\}$$

for every $S \subseteq M$ and $e \in M \setminus S$.

Hence, negative dichotomous valuations and dichotomous cost functions are equivalent under the transformation $c_i = -v_i$. In particular, every c_i is nondecreasing and integer-valued. We use the valuation and cost representations interchangeably.

2.2 Fairness with subsidies

Definition 2.3 (Envy-freeness). An allocation $\mathcal{A}$ is *envy-free* (EF) if, for all $i, j \in N$,

$$c_i(A_i) \leq c_i(A_j).$$

A subsidy vector is $p = (p_1, \dots, p_n) \in \mathbb{R}_+^n$.

Definition 2.4 (Envy-free solution). An allocation $\mathcal{A}$ together with a subsidy vector p forms an *envy-free solution* if

$$c_i(A_i) - p_i \leq c_i(A_j) - p_j \quad \text{for all } i, j \in N.$$

An allocation is *envy-freeable* if it admits some nonnegative subsidy vector that makes it envy-free. Halpern and Shah [12] characterize envy-freeable allocations and their pointwise-minimal

supporting subsidies; our construction below instead specifies both the allocation and a supporting subsidy vector directly.

Since every item in our setting is a chore, we use the standard chore version of envy-freeness up to one item.

Definition 2.5 (EF1 for chores). A complete allocation $\mathcal{A}$ is *EF1* if, for every $i, j \in N$, either

$$c_i(A_i) \leq c_i(A_j),$$

or there exists a chore $e \in A_i$ such that

$$c_i(A_i \setminus \{e\}) \leq c_i(A_j).$$

For all-chore instances, this is equivalent to the usual two-sided definition of EF1: if removing an item from the envied bundle eliminates envy, then the original allocation was already envy-free for that pair.

3 Unit Subsidies for Negative Dichotomous Valuations

This section proves the main theorem: for any number of agents n and every negative dichotomous instance, there is an allocation admitting an envy-free solution with a subsidy of 0 or 1 per agent, at most $n - 1$ in total, computable in polynomial time.

Our argument starts from an envy-free *partial* allocation produced by the algorithm of Tao, Wu, Yu, and Zhou [18], and then completes this partial allocation. We recall in Section 3.1 the relevant terminal-state properties of their algorithm, including Theorem 6.1 and the three update rules used in its construction. These ingredients are due to Tao et al. [18]. The completion procedure introduced in Section 3.2, together with the subsequent lemmas and theorems, constitutes our contribution.

We work in cost form throughout, so an envy-free solution (Definition 2.4) is the requirement $(\mathcal{A}, p)$ such that

$$c_i(A_i) - p_i \leq c_i(A_j) - p_j \quad \text{for all } i, j \in N, \quad (1)$$

and a partial allocation X is envy-free (Definition 2.3) when

$$c_i(X_i) \leq c_i(X_j) \quad \text{for all } i, j \in N. \quad (2)$$

For $S \subseteq M$ and $e \in M \setminus S$ write $c_i(e \mid S) := c_i(S \cup \{e\}) - c_i(S) \in \{0, 1\}$ for a marginal, and for a partial allocation X write $R(X) := M \setminus \bigcup_{i \in N} X_i$ for its *residue*. Moreover, as c_i s are integer valued, for all $S, T \subseteq M$, $c_i(S) < c_i(T) \implies c_i(S) \leq c_i(T) - 1$.

To describe the algorithm of Tao, Wu, Yu, and Zhou [18] and our completion procedure, we use the following standard graph-theoretic notions.

Definition 3.1 (Strongly connected component and tail component). Let $G = (V, E)$ be a finite directed graph. Two vertices $u, v \in V$ are *strongly connected* if either $u = v$, or there is a directed path from u to v and a directed path from v to u . Strong connectivity is an equivalence relation on V , and its equivalence classes are the *strongly connected components* of G .

A strongly connected component S is a *tail component* (or *sink component*) if no arc leaves S ; that is, there is no $(i, j) \in E$ with $i \in S$ and $j \in V \setminus S$.

The following standard graph-theoretic fact will be used repeatedly.

LEMMA 3.2. *Every finite non-empty directed graph has at least one tail strongly connected component.*

PROOF. Start with any strongly connected component. If it is not a tail component, follow an outgoing arc to another strongly connected component and repeat. No component can be visited twice, since this would make the two components mutually reachable. As there are finitely many components, the process terminates at a component with no outgoing arc, which is a tail component. $\square$

3.1 The terminal state of the partial-allocation algorithm

THEOREM 3.3 ([18], THEOREM 6.1). *For every n and all dichotomous cost functions $c_1, \dots, c_n$ there is a polynomial-time algorithm returning a partial allocation X that is envy-free in the sense of (2) and satisfies $|R(X)| \leq n - 1$.*

The proof of our main result uses not only the statement of Theorem 3.3, but also structural properties of the terminal state of the algorithm of [18]. We therefore recall the relevant part of that algorithm. It maintains an envy-free partial allocation $X = (X_1, \dots, X_n)$ together with the *equality graph*

$$G = (N, E), \quad (i, j) \in E \iff i \neq j \text{ and } c_i(X_i) = c_i(X_j). \quad (3)$$

While $R(X) \neq \emptyset$, the algorithm considers the following rules in the stated order.

- (R1) If there exist $e \in R(X)$ and $i \in N$ such that $c_i(e \mid X_i) = 0$, assign e to i .
- (R2) Otherwise, if there exist $e \in R(X)$ and an arc $(i, j) \in E$ lying on a directed cycle of G such that $c_i(e \mid X_j) = 0$, rotate the bundles along that cycle and assign e to i .
- (R3) Otherwise, select a tail strongly connected component S of G . If $|R(X)| \geq |S|$, assign distinct residual chores to the agents of S , one chore per agent. Otherwise, halt.

Consequently, if the algorithm terminates with $R(X) \neq \emptyset$, then neither (R1) nor (R2) is applicable at the terminal state. Moreover, if S denotes the tail strongly connected component selected in the terminating application of (R3), then

$$|R(X)| < |S|.$$

For the remainder of this section, let X denote the terminal partial allocation of the algorithm, and write

$$R := R(X), \quad r := |R|.$$

Unless stated otherwise, let $G = (N, E)$ denote the equality graph defined in (3) at this terminal state. When $r \geq 1$, let S be the tail SCC selected at the terminating application of (R3); therefore $r < |S|$.

LEMMA 3.4. *Suppose $r \geq 1$. Then*

- (i) $c_i(e \mid X_i) = 1$ for every $i \in N$ and every $e \in R$;
- (ii) $c_i(e \mid X_j) = 1$ for every arc $(i, j) \in E$ with $i, j \in S$ and every $e \in R$;
- (iii) $c_i(X_i) < c_i(X_j)$ for every $i \in S$ and every $j \in N \setminus S$.

PROOF. Since $r \geq 1$, the algorithm terminates with a nonempty residue. Therefore, neither (R1) nor (R2) is applicable at the terminal state.

(i) If $c_i(e \mid X_i) = 0$ for some $i \in N$ and $e \in R$, then (R1) would be applicable. Since every marginal cost belongs to $\{0, 1\}$ by Definition 2.2, it follows that

$$c_i(e \mid X_i) = 1.$$

(ii) Let $(i, j) \in E$ with $i, j \in S$. Since S is strongly connected, there is a directed path from j to i . Choosing such a path with no repeated vertices and appending the arc (i, j) yields a directed cycle containing (i, j) . If $c_i(e \mid X_j) = 0$ for some $e \in R$, then (R2) would be applicable to this arc and chore. Hence $c_i(e \mid X_j) \neq 0$, and therefore

$$c_i(e \mid X_j) = 1.$$

(iii) Let $i \in S$ and $j \in N \setminus S$. Since S is a tail strongly connected component, no arc leaves S ; in particular, $(i, j) \notin E$. By (3),

$$c_i(X_i) \neq c_i(X_j).$$

On the other hand, envy-freeness of X , as expressed in (2), gives

$$c_i(X_i) \leq c_i(X_j).$$

The inequality must therefore be strict. $\square$

3.2 The completion

Assume $r \geq 1$ and write $R = \{e_1, \dots, e_r\}$. Since $r < |S|$, we may choose r distinct agents $T = \{t_1, \dots, t_r\} \subseteq S$ arbitrarily. Define the allocation $\mathcal{A} = (A_1, \dots, A_n)$ by

$$A_i := \begin{cases} X_{t_k} \cup \{e_k\}, & i = t_k \quad (1 \leq k \leq r), \\ X_i, & i \in N \setminus T. \end{cases} \quad (4)$$

Since the agents $t_1, \dots, t_r$ are distinct and each chore in R is assigned exactly once, the bundles in $\mathcal{A}$ remain pairwise disjoint and their union is M . Hence, $\mathcal{A}$ is a complete allocation. We refer to the agents in T as the *recipients*.

We next construct a subsidy vector $p \in \{0, 1\}^n$ and show that $(\mathcal{A}, p)$ is envy-free. The following definition specifies the set of agents who receive one unit of subsidy.

Definition 3.5 (*Subsidy set*). Let $P \subseteq N$ be the smallest set satisfying

- (P1) $T \subseteq P$; and
- (P2) if $i \in N \setminus S$ and $(i, j) \in E$ for some $j \in P$, then $i \in P$.

Equivalently, initialise $P := T$ and repeatedly add any agent $i \in N \setminus S$ for which $(i, j) \in E$ for some j already in P . Continue until no further agents can be added. Since N is finite and the set P only grows, this procedure terminates. Define

$$p_i := \begin{cases} 1, & i \in P, \\ 0, & i \in N \setminus P. \end{cases} \quad (5)$$

The complete procedure is summarised in Algorithm 1. The partial-allocation phase in Line 2 is the partial-allocation algorithm of Tao et al. [18]; the completion and subsidy construction in the subsequent lines constitute our contribution.

Algorithm 1 EF1 allocation with at most one unit of subsidy per agent for negative dichotomous valuations

Require: Agents $N = [n]$, chores M , and value-oracle access to negative dichotomous valuations $\{v_i\}_{i \in N}$.

Ensure: A complete allocation $\mathcal{A}$ that is EF1, together with a subsidy vector $p \in \{0, 1\}^n$ such that $(\mathcal{A}, p)$ is envy-free and $\sum_{i \in N} p_i \leq n - 1$.

- 1: Set $c_i = -v_i$ for every $i \in N$.
- 2: Run the partial-allocation algorithm of Tao et al. [18] on $\{c_i\}_{i \in N}$, retaining the tail strongly connected component selected in the terminating application of (R3) if the residue is nonempty.
- 3: Let $X = (X_1, \dots, X_n)$ be its terminal partial allocation and let

$$R = M \setminus \bigcup_{i \in N} X_i.$$

- 4: **if** $R = \emptyset$ **then**
- 5: **return** $(X, 0)$.
- 6: Construct the terminal equality graph $G = (N, E)$, where

$$(i, j) \in E \iff i \neq j \text{ and } c_i(X_i) = c_i(X_j).$$
- 7: Write $R = \{e_1, \dots, e_r\}$.
- 8: Let S be the tail strongly connected component selected in the terminating application of (R3). $\triangleright r < |S|$
- 9: Choose arbitrary distinct agents $T = \{t_1, \dots, t_r\} \subseteq S$. $\triangleright$
possible since $r < |S|$
- 10: **for** $k = 1, \dots, r$ **do**
- 11: $A_{t_k} \leftarrow X_{t_k} \cup \{e_k\}$.
- 12: **for** $i \in N \setminus T$ **do**
- 13: $A_i \leftarrow X_i$.
- 14: Initialise $P \leftarrow T$.
- 15: **while** there exist $i \in N \setminus S$ and $j \in P$ with $(i, j) \in E$ and $i \notin P$ **do**
- 16: $P \leftarrow P \cup \{i\}$.
- 17: **for** $i \in N$ **do**
- 18: $p_i \leftarrow \mathbf{1}[i \in P]$.
- 19: $\mathcal{A} \leftarrow (A_1, \dots, A_n)$.
- 20: **return** $(\mathcal{A}, p)$.

LEMMA 3.6. $P \cap S = T$. Equivalently,

$$P \setminus T \subseteq N \setminus S.$$

Thus, every subsidised agent in S is a recipient.

PROOF. Since $T \subseteq P$ by (P1) and $T \subseteq S$, we have $T \subseteq P \cap S$. Suppose, for contradiction, that there exists $i \in (P \cap S) \setminus T$. Consider

$$P' := P \setminus \{i\}.$$

Since $i \notin T$, the set P' still satisfies (P1). We claim that P' also satisfies (P2). Let $k \in N \setminus S$ and suppose that $(k, j) \in E$ for some $j \in P'$. Since $P' \subseteq P$ and P satisfies (P2), we have $k \in P$. Moreover, $k \neq i$, because $k \notin S$ whereas $i \in S$. Hence $k \in P'$.

Thus, P' satisfies both (P1) and (P2), contradicting the minimality of P . Therefore $P \cap S = T$. $\square$

By (5), $p \in \{0, 1\}^n$. Moreover, $|T| = r < |S|$, so $S \setminus T \neq \emptyset$. Since Lemma 3.6 gives $P \cap S = T$, every agent in $S \setminus T$ lies outside P and

hence receives zero subsidy. Therefore,

$$\sum_{i \in N} p_i \leq n - 1. \quad (6)$$

LEMMA 3.7. Let $i \in S$, $t \in T$, and $e \in R$. Then

$$c_i(X_t \cup \{e\}) \geq c_i(X_i) + 1.$$

PROOF. Since X is envy-free, $c_i(X_i) \leq c_i(X_t)$. If the inequality is strict, the integrality of the cost functions gives $c_i(X_t) \geq c_i(X_i) + 1$, and the desired inequality follows from the monotonicity of c_i . Suppose instead that $c_i(X_i) = c_i(X_t)$. If $i = t$, then Lemma 3.4(i) gives

$$c_i(e \mid X_i) = 1.$$

If $i \neq t$, then $(i, t) \in E$ by (3); since $i, t \in S$, Lemma 3.4(ii) gives

$$c_i(e \mid X_t) = 1.$$

In either case,

$$c_i(X_t \cup \{e\}) = c_i(X_i) + 1.$$

$\square$

3.3 Envy-freeness of the completion

THEOREM 3.8. Let X , R , and S be as in Section 3.1 with $r \geq 1$, let $\mathcal{A}$ be defined by (4), and let p be defined by (5). Then $(\mathcal{A}, p)$ is an envy-free solution.

PROOF. Fix $i \in N$. Since $X_j \subseteq A_j$ for every $j \in N$, monotonicity of c_i and envy-freeness of X give

$$c_i(A_j) \geq c_i(X_j) \geq c_i(X_i) \quad \text{for every } j \in N. \quad (7)$$

We verify $c_i(A_i) - p_i \leq c_i(A_j) - p_j$ for every $j \in N$.

Case 1: $i \in T$. Then $p_i = 1$, and if $e \in R$ is the residual chore assigned to i , Lemma 3.4(i) gives

$$c_i(A_i) = c_i(X_i) + c_i(e \mid X_i) = c_i(X_i) + 1.$$

Hence $c_i(A_i) - p_i = c_i(X_i)$. If $j \notin P$, then $j \notin T$, so $A_j = X_j$ and $p_j = 0$. Thus, by envy-freeness of X ,

$$c_i(A_j) - p_j = c_i(X_j) \geq c_i(X_i).$$

Suppose instead that $j \in P$. If $j \in T$, then Lemma 3.7 gives $c_i(A_j) \geq c_i(X_j) + 1$. If $j \in P \setminus T$, then Lemma 3.6 implies $j \notin S$. Since $i \in S$, Lemma 3.4(iii) gives $c_i(X_i) < c_i(X_j)$. By integrality, $c_i(A_j) = c_i(X_j) \geq c_i(X_i) + 1$. Thus, in either case, since $p_j = 1$,

$$c_i(A_j) - p_j \geq c_i(X_i).$$

Case 2: $i \in P \setminus T$. Here $A_i = X_i$ and $p_i = 1$, so $c_i(A_i) - p_i = c_i(X_i) - 1$. For every $j \in N$, (7) and $p_j \leq 1$ imply

$$c_i(A_j) - p_j \geq c_i(X_i) - 1 = c_i(A_i) - p_i.$$

Case 3: $i \notin P$. Since $T \subseteq P$, we have $i \notin T$, and therefore $A_i = X_i$ and $p_i = 0$. Hence $c_i(A_i) - p_i = c_i(X_i)$. If $j \notin P$, then $A_j = X_j$ and $p_j = 0$, and envy-freeness of X gives

$$c_i(A_j) - p_j = c_i(X_j) \geq c_i(X_i) = c_i(A_i) - p_i.$$

Now suppose that $j \in P$. We claim that

$$c_i(A_j) \geq c_i(X_i) + 1. \quad (8)$$

If $i \notin S$, envy-freeness of X gives $c_i(X_i) \leq c_i(X_j)$. Equality would imply $(i, j) \in E$ by (3); since $i \in N \setminus S$ and $j \in P$, condition (P2)

would then imply $i \in P$, a contradiction. Hence $c_i(X_i) < c_i(X_j)$, and integrality together with monotonicity yield

$$c_i(A_j) \geq c_i(X_j) \geq c_i(X_i) + 1.$$

It remains to consider $i \in S$. If $j \in T$, then (8) follows from Lemma 3.7. If $j \in P \setminus T$, then Lemma 3.6 gives $j \notin S$, and Lemma 3.4(iii), together with integrality, gives

$$c_i(A_j) = c_i(X_j) \geq c_i(X_i) + 1.$$

Thus (8) holds in all cases. Since $p_j = 1$,

$$c_i(A_j) - p_j \geq c_i(X_i) = c_i(A_i) - p_i.$$

The envy-freeness inequality therefore holds for every pair $i, j \in N$. $\square$

3.4 The EF1 property of the completion

COROLLARY 3.9. *The complete allocation $\mathcal{A}$ constructed in Section 3.2 is EF1 in the sense of Definition 2.5. More precisely:*

- (i) every non-recipient $i \in N \setminus T$ is envy-free in $\mathcal{A}$; and
- (ii) for every recipient $t_k \in T$,

$$c_{t_k}(A_{t_k} \setminus \{e_k\}) \leq c_{t_k}(A_j) \quad \text{for every } j \in N.$$

PROOF. Recall that the partial allocation $X = (X_1, \dots, X_n)$ is envy-free and that $X_j \subseteq A_j$ for every $j \in N$. Let first $i \in N \setminus T$. Then $A_i = X_i$. Hence, for every $j \in N$, envy-freeness of X and monotonicity of c_i give

$$c_i(A_i) = c_i(X_i) \leq c_i(X_j) \leq c_i(A_j).$$

Thus, every non-recipient is envy-free in $\mathcal{A}$. Now let $i = t_k \in T$. By construction,

$$A_i = X_i \cup \{e_k\}, \quad \text{and hence} \quad A_i \setminus \{e_k\} = X_i.$$

Therefore, for every $j \in N$,

$$c_i(A_i \setminus \{e_k\}) = c_i(X_i) \leq c_i(X_j) \leq c_i(A_j),$$

where the two inequalities follow from envy-freeness of X and monotonicity of c_i , respectively. Thus, removing the residual chore assigned to a recipient eliminates all of her envy. Consequently, $\mathcal{A}$ is EF1. $\square$

3.5 The main theorem

THEOREM 3.10. *For every instance $\langle N, M, \{v_i\}_{i \in N} \rangle$ with negative dichotomous valuations, there exist a complete allocation $\mathcal{A}$ and a subsidy vector $p \in \{0, 1\}^n$ such that*

$$\sum_{i \in N} p_i \leq n - 1$$

and $(\mathcal{A}, p)$ is an envy-free solution. Moreover, $\mathcal{A}$ is EF1 even before subsidies are paid. Such an allocation and subsidy vector can be computed in polynomial time given value-oracle access.

PROOF. Pass to the cost representation $c_i = -v_i$, which is dichotomous in the sense of Definition 2.2. Run the partial-allocation algorithm of Tao et al. [18], and let X be its terminal partial allocation, with residue

$$R := R(X), \quad r := |R|.$$

If $r = 0$, then X is complete and satisfies (2). Hence $(X, 0)$ is an envy-free solution. Since every envy-free allocation is EF1, X is also EF1.

If $r \geq 1$, let $G = (N, E)$ be the terminal equality graph and let S be the tail strongly connected component of G fixed in Section 3.1. Construct T , $\mathcal{A}$, and p as in Section 3.2. Then $\mathcal{A}$ is complete, Theorem 3.8 shows that $(\mathcal{A}, p)$ is envy-free, (5) and (6) give

$$p \in \{0, 1\}^n \quad \text{and} \quad \sum_{i \in N} p_i \leq n - 1,$$

and Corollary 3.9 shows that $\mathcal{A}$ is EF1.

It remains to verify polynomial-time computability. The partial-allocation algorithm of Tao et al. [18] runs in polynomial time. If $r \geq 1$, the terminal equality graph has n vertices and can be constructed from the n^2 values $c_i(X_j)$, requiring one value-oracle query for each pair (i, j) . Its strongly connected components can be computed in polynomial time, after which a tail component S is obtained by selecting one with no outgoing arc. By Theorem 3.3, $r \leq n - 1$, so constructing $\mathcal{A}$ requires at most $n - 1$ additional assignments. Finally, P can be computed by a backward reachability search from T , restricted to vertices outside S , in $O(n^2)$ time. Hence, the entire procedure runs in polynomial time. $\square$

As noted in the introduction, the total-subsidy bound in Theorem 3.10 is tight: there are instances for which every complete envy-free solution requires total subsidy at least $n - 1$.

3.6 Pareto optimality and subsidies

In this section, we show that the unit-subsidy guarantee need not be compatible with Pareto optimality of the underlying allocation. Pareto optimality is evaluated with respect to the allocation of chores, without taking subsidies into account. Formally, a complete allocation $\mathcal{B}$ *Pareto dominates* a complete allocation $\mathcal{A}$ if

$$c_i(\mathcal{B}_i) \leq c_i(\mathcal{A}_i) \quad \text{for every } i \in N,$$

with strict inequality for at least one agent. A complete allocation is *Pareto optimal* if it is not Pareto dominated by any other complete allocation.

PROPOSITION 3.11. *There exist negative dichotomous instances, even with two agents and identical binary submodular cost functions, for which no Pareto-optimal allocation admits an envy-free subsidy vector in $\{0, 1\}^n$.*

PROOF. Consider two agents $N = \{1, 2\}$ and three chores $M = \{a, b, c\}$. Both agents have the same cost function

$$c_1(S) = c_2(S) = \min\{|S|, 2\}.$$

Every marginal cost belongs to $\{0, 1\}$, so the cost function is dichotomous. Moreover, it is submodular, since $k \mapsto \min\{k, 2\}$ is a nondecreasing concave function of cardinality. It is not additive, since

$$c_i(\{a, b, c\}) = 2 < 3 = \sum_{g \in \{a, b, c\}} c_i(\{g\}).$$

Every complete allocation has, up to symmetry, either bundle-size profile $(3, 0)$ or $(2, 1)$. First, consider the allocation in which agent 1 receives all three chores. Its cost profile is $(2, 0)$. The possible cost profiles of complete allocations are

$$(2, 0), \quad (2, 1), \quad (1, 2), \quad (0, 2),$$

and none of the latter three Pareto dominates $(2, 0)$. Hence, the concentrated allocation is Pareto optimal. By symmetry, the allocation in which agent 2 receives all three chores is also Pareto optimal. However, a concentrated allocation cannot be made envy-free with unit subsidies. If agent 1 receives all three chores, envy-freeness requires

$$2 - p_1 \leq -p_2,$$

and hence

$$p_1 - p_2 \geq 2.$$

Thus, no subsidy vector in $\{0, 1\}^2$ suffices. The case in which agent 2 receives all three chores is symmetric.

Now consider a split allocation, with bundle sizes $(2, 1)$ up to symmetry. Its cost profile is $(2, 1)$, so it can be made envy-free with subsidies $(1, 0)$, up to symmetry. Nevertheless, it is not Pareto optimal. Assigning the remaining chore to the agent who already holds two chores changes the cost profile from $(2, 1)$ to $(2, 0)$: the cost of that agent remains unchanged, while the other agent's cost strictly decreases.

Thus, the Pareto-optimal allocations are precisely the two concentrated allocations, and each requires a subsidy of at least 2 for the agent receiving all three chores. Conversely, every allocation that admits an envy-free subsidy vector in $\{0, 1\}^2$ is Pareto dominated. Therefore, no allocation is simultaneously Pareto optimal and envy-free with subsidies in $\{0, 1\}^2$. $\square$

Thus, the unit-subsidy guarantee of Theorem 3.10 cannot, in general, be strengthened to require Pareto optimality.

A Failure of an Incremental Unit-Subsidy Extension

Example A.1 (Failure of a BKNS-type incremental extension). Consider three agents $N = \{1, 2, 3\}$ and three chores $M = \{x, y, z\}$. We work in cost form, writing $c_i = -v_i$ for agent i 's non-negative cost function. Thus, an allocation $\mathcal{A} = (A_1, \dots, A_n)$ together with a subsidy vector $p = (p_1, \dots, p_n)$ is envy-free iff

$$c_i(A_i) - p_i \leq c_i(A_j) - p_j \quad \text{for all } i, j \in N.$$

The cost functions depend only on bundle size:

$$c_1(S) = \max\{0, |S| - 1\}, \quad c_2(S) = c_3(S) = \min\{|S|, 2\}.$$

Thus,

$ S $	0	1	2	3
$c_1(S)$	0	0	1	2
$c_2(S) = c_3(S)$	0	1	2	2

Every marginal cost belongs to $\{0, 1\}$, so these are dichotomous cost functions. Suppose z is currently unallocated, and the partial allocation is

$$X_1 = \{x, y\}, \quad X_2 = X_3 = \emptyset.$$

Note that this partial allocation is itself reachable from the empty allocation through successive insertions while maintaining the unit-subsidy invariant:

$$(\emptyset, \emptyset, \emptyset) \longrightarrow (\{x\}, \emptyset, \emptyset) \longrightarrow (\{x, y\}, \emptyset, \emptyset),$$

with pointwise-minimal subsidy vectors $(0, 0, 0)$, $(0, 0, 0)$, and $(1, 0, 0)$, respectively. The pointwise-minimal subsidy vector of (X_1, X_2, X_3) is

$$p^* = (1, 0, 0).$$

Indeed, agent 1 has cost 1 for her own bundle and cost 0 for either empty bundle, so she requires one unit of subsidy. Agents 2 and 3 have no envy, since they have cost 0 for their own empty bundles and cost 2 for X_1 .

We claim that the remaining chore z cannot be inserted so that the resulting allocation admits an envy-free subsidy vector in $\{0, 1\}^3$, even if the existing bundles $\{x, y\}, \emptyset, \emptyset$ may first be permuted among the agents.

If z is added to the two-chore bundle, the resulting bundle sizes are $(3, 0, 0)$. Every agent evaluates a three-chore bundle at cost 2 and an empty bundle at cost 0. Hence, the agent receiving the three-chore bundle requires a subsidy at least two units larger than that of an agent receiving an empty bundle. Thus, no subsidy vector in $\{0, 1\}^3$ can make the resulting allocation envy-free.

It remains to consider the case in which z is added to an empty bundle. The resulting bundle sizes are $(2, 1, 0)$, up to permutation. If an agent in $\{2, 3\}$ receives the two-chore bundle, she evaluates it at cost 2 and the empty bundle at cost 0, again requiring a subsidy difference of at least two.

Therefore, the only remaining possibility is that agent 1 receives the two-chore bundle. Let $i \in \{2, 3\}$ receive the singleton and let j receive the empty bundle. Envy-freeness for agent 1 with respect to i requires

$$1 - p_1 \leq 0 - p_i,$$

and hence

$$p_1 \geq p_i + 1.$$

Envy-freeness for agent i with respect to j requires

$$1 - p_i \leq -p_j,$$

and hence

$$p_i \geq p_j + 1.$$

Combining the two inequalities gives

$$p_1 \geq p_j + 2,$$

which is impossible for $p \in \{0, 1\}^3$. Hence, no assignment of the additional chore z , even after an arbitrary permutation of the previously allocated bundles, preserves the unit-subsidy guarantee. Notice that this is an obstruction to the incremental construction, not to the instance itself. The complete allocation

$$(\{x\}, \{y\}, \{z\})$$

is envy-free without subsidies: agent 1 evaluates every singleton at cost 0, while agents 2 and 3 evaluate every singleton at cost 1. $\square$

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